

Structured Financial Complaint Analysis with Hybrid Retrieval and Threshold-Based Abstention: A Reproducible NLP Pipeline

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Abstract—Consumer complaints submitted to financial regulators are a useful source of operational insight, but their unstructured and heavily redacted form makes automated processing difficult to scale. Although Large Language Models (LLMs) can assist with summarization, their outputs may contain unsupported or factually incorrect content, which is particularly risky in regulated settings. In this paper, we describe a modular NLP pipeline for structured financial complaint analysis that combines a redaction-aware preprocessor, a fine-tuned DistilBERT classifier, hybrid BM25 and dense retrieval with Reciprocal Rank Fusion (RRF), and a confidence-based human-review gate.

On a stratified split of 34,999 training, 7,501 validation, and 7,500 test records drawn from a 50,000-record sample of the CFPB complaint corpus, DistilBERT achieves 87.60% accuracy and a weighted F1-score of 0.8709, outperforming the evaluated TF-IDF baselines. The hybrid retrieval stage achieves a Precision@5 of 0.820 under a product-label match proxy and 0.521 under a stricter product-plus-issue match proxy. A fixed raw-softmax threshold of 0.60 flags 13.33% (4/30) of cases for human review in a 30-sample mock-generation evaluation.

The generation stage was evaluated only in deterministic mock mode; therefore, the reported results do not establish LLM factuality, retrieval faithfulness, or resolution-drafting quality. All reported experiments use the 50,000-record sample, while the repository includes chunked ingestion support for the larger CFPB CSV. The results indicate that preprocessing, contextual classification, and hybrid retrieval provide a practical foundation for structured complaint analysis, while calibration, human relevance judgments, and full LLM evaluation remain future work.

I. INTRODUCTION

The modern financial ecosystem depends heavily on regulatory bodies to ensure fair lending practices, market transparency, and consumer protection. The United States Consumer Financial Protection Bureau (CFPB) maintains an extensive public database of consumer disputes, detailing adverse scenarios ranging from unauthorized credit card charges to complex mortgage escrow discrepancies [1]. These narratives serve as a vital feedback mechanism but present a significant computational challenge: extracting actionable operational intelligence from raw, unstructured, and highly anonymized text. Many complaints are also written in forms where simple keyword matching is insufficient, particularly when the narrative is short, heavily redacted, or expressed through varied terminology.

Historically, complaint resolution has operated as a manual triage process. Analysts must read narratives sequentially, cross-reference grievances against complex internal regulatory rubrics, identify historically similar resolutions to ensure legal parity, and draft corrective summaries. As digital transaction

volumes expand, this labor-intensive approach is fundamentally unscalable, creating massive operational bottlenecks.

Recent advancements in Natural Language Processing (NLP) offer pathways to automation. However, deploying generic, autonomous Large Language Models (LLMs) in finance introduces severe systemic risks. Unconstrained generative architectures suffer from hallucinations—the generation of factually incorrect or unverified information [2]—and typically fail to assign calibrated, reliable confidence scores to their outputs [3]. In regulatory compliance, confidently generating an incorrect legal resolution is far more damaging than abstaining from a decision entirely.

Research Gap: Current methodologies largely bifurcate into two incomplete approaches. First, strictly deterministic classification systems categorize complaints accurately but fail to synthesize actionable context or draft resolutions. Second, purely generative Retrieval-Augmented Generation (RAG) pipelines synthesize fluid text but lack mathematical guardrails and explicit abstention mechanisms tailored for high-risk domains, forcing automated answers even on out-of-distribution inputs.

Our Contribution: This study introduces a modular, computationally rigorous pipeline designed specifically for financial texts. The architecture replaces monolithic generation with a four-stage process, fully verifiable in our repository codebase:

1. **Redaction-Aware Preprocessing:** Dynamically filtering anomalous anonymization masks natively present in the CFPB corpus to preserve semantic structures.

2. **Transformer-Based Classification:** Utilizing a fine-tuned distilbert-base-uncased model to accurately map narratives to standardized financial products, while improving performance across the imbalanced class distribution.

3. **Hybrid Information Retrieval:** Running sparse BM25 retrieval and dense all-MiniLM-L6-v2 embeddings via FAISS in parallel, followed by Reciprocal Rank Fusion (RRF) to retrieve related records from the training corpus. Retrieval quality is evaluated using label-match proxy criteria.

4. **Threshold-Based Selective Abstention:** Applying a fixed confidence threshold to classifier outputs and routing low-confidence cases to human review.

The main idea is not to let one model do everything. Instead, the pipeline separates preprocessing, classification, retrieval, and structured generation, while exposing classifier confidence and an explicit human-review flag for lower-confidence cases.

II. RELATED WORK

The application of computational linguistics to regulatory data spans classical text categorization, deep-learning embeddings, retrieval architectures, and controlled generation.

A. Classical Text Classification: Early machine learning models applied to text corpora relied on Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) feature engineering. Foundational research by Joachims [4] established that TF-IDF modeling combined with Support Vector Machines (SVM) or Logistic Regression yields robust baselines for text categorization. However, linear classifiers treat text as a static, unordered set. They inherently fail to parse sequential dependencies, complex syntactic negations, and the contextual nuances prevalent in lengthy legal or financial complaints, often collapsing when faced with vocabulary outside their training distribution.

B. Transformer Architectures in Finance: The introduction of the Transformer architecture [5] and BERT (Bidirectional Encoder Representations from Transformers) [6] revolutionized semantic representation. By leveraging self-attention mechanisms, Transformers generate deeply contextualized token embeddings. To mitigate the computational overhead of processing large financial datasets, Sanh et al. introduced DistilBERT [7], utilizing knowledge distillation to retain 97% of BERT’s language understanding while reducing parameters by 40%. Domain-specific adaptations like FinBERT [8] have further demonstrated that pre-trained language models significantly outperform classical baselines on financial sentiment and classification tasks, largely due to their ability to comprehend complex financial jargon.

C. Retrieval-Augmented Generation (RAG) and Hybrid Search: To address the hallucination tendencies of parametric generative models, Lewis et al. introduced RAG [9], which augments neural networks with a non-parametric factual database. Effective retrieval in RAG necessitates exceptionally high precision. While dense embeddings—such as SentenceBERT [10] coupled with efficient nearest-neighbor searches like FAISS [11]—excel at identifying broad semantic similarity, they often miss exact keyword matches crucial for rigid regulatory codes. Conversely, sparse models like BM25 [12] mathematically prioritize exact term overlap but miss paraphrased intent. Reciprocal Rank Fusion (RRF) [13] is now frequently employed to normalize and merge these contrasting modalities, ensuring both broad semantic intent and explicit terminology are captured simultaneously.

D. Selective Classification and Confidence-Based Abstention: As autonomous systems become larger, it is increasingly important to recognize their limits. Expected Calibration Error (ECE) measures the gap between predicted confidence and empirical correctness [14], while selective prediction allows a classifier to abstain on inputs for which its confidence is insufficient [15]. These ideas are related but distinct: a confidence threshold can provide a simple abstention mechanism without establishing that predicted probabilities are calibrated.

In this work, we use a fixed threshold on the classifier’s raw softmax confidence as a routing mechanism for human review; probability calibration and ECE measurement are left for future work.

III. PROPOSED METHODOLOGY

To eliminate the vulnerabilities of unchecked text generation, this research employs a rigidly modular pipeline. The methodology transitions sequentially from raw data handling to deep representation learning, hybrid search algorithms, and structured generation with explicit constraints.

A. Data Preprocessing and Stratification: Consumer data from the CFPB contains aggressive redaction markers designed to anonymize personally identifiable information. If left untreated, these markers act as massive statistical noise for classical classifiers. During preprocessing, raw text strings T_i undergo localized transformations to maximize the signal-to-noise ratio: 1. **Redaction Purging:** A deterministic regular expression matches contiguous anonymization tokens (e.g., XXXX, XX/XX/XXXX) and collapses them into single space characters. This prevents feature explosion in TF-IDF matrices and reduces meaningless token consumption in Transformers. In preliminary experiments, retaining these placeholders caused sparse representations to allocate disproportionate capacity to anonymization tokens rather than complaint content, motivating their deterministic removal. 2. **Length Normalization:** Strings are lowercased and internal whitespace is collapsed. To accommodate contextual window limits and filter garbage data, texts are strictly truncated to a 5,000-character maximum. Narratives shorter than 50 characters—often the result of heavy redaction leaving no semantic value—are dropped entirely.

To keep the process deterministic and reproducible, a fixed random seed of 42 is used to sample up to 50,000 records from the preprocessed corpus, followed by a stratified split across 10 target financial product classes. The resulting dataset contains 34,999 training, 7,501 validation, and 7,500 test records. The resulting class distribution remains highly imbalanced because stratification preserves the observed class proportions rather than equalizing class counts.

All reported metrics in this paper are computed on this 50,000-record sample. The repository includes chunked loading support for the larger CFPB CSV, but the experiments reported here were not run on the full corpus. The validation partition is held out from the reported training and test metrics and is retained for future threshold-tuning and calibration experiments.

B. Classification Models: We map each complaint x_i to a discrete financial product label $y_i \in \{1, \dots, C\}$ where $C = 10$.

1) Classical Baselines: TF-IDF vectors (configured to a maximum of 50,000 features using uni/bi-grams) are extracted. Our Logistic Regression implementation optimizes the Cross-Entropy loss using the L-BFGS solver. A Linear Support Vector Machine is also evaluated, utilizing Platt scaling [16] to convert geometric margins $f(x)$ into probabilistic outputs. A Multinomial Naive Bayes model is included as a fundamental benchmark.

System Architecture of the CFPB Complaint-to-Resolution Pipeline

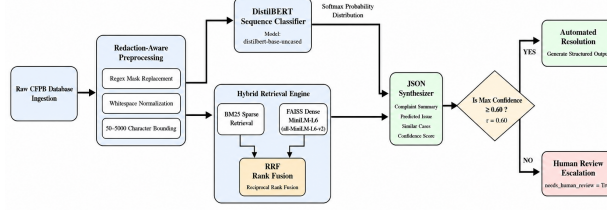


Figure 1: System architecture of the CFPB complaint-to-resolution pipeline.

2) Fine-Tuned DistilBERT Classifier: The primary analytical engine utilizes the distilbert-base-uncased architecture, which applies multi-head dot-product attention [5]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

A linear classification head is appended to the final hidden state of the [CLS] token. Logits z_i are passed through a softmax function to produce class probability distributions:

$$\text{Conf}_{\text{cls}}(y_i | x) = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)} \quad (2)$$

The model is fine-tuned using the AdamW optimizer with a learning rate of 2.0×10^{-5} , a weight decay of 0.01, and a linear learning rate scheduler incorporating 100 warmup steps over 3 epochs. The maximum sequence length is set to 192 tokens with a batch size of 8 to accommodate local GPU constraints (4GB VRAM).

C. Hybrid Information Retrieval: To provide contextual examples for downstream structured synthesis, the pipeline retrieves the top- k related records from the training corpus for a given query q .

1) Sparse Retrieval (BM25): BM25 captures exact textual overlap for specific financial terminology [12]:

$$\text{score}_{\text{BM25}}(D, q) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, D) (k_1 + 1)}{f(t, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \quad (3)$$

2) Dense Retrieval (FAISS): Queries are embedded using sentence-transformers/all-MiniLM-L6-v2 to produce dense $d = 384$ dimensional vectors $\vec{v}_q$. The historical corpus is pre-indexed using FAISS (Facebook AI Similarity Search) [11] to efficiently compute exact Cosine Similarities:

$$\text{Cosine Similarity}(\vec{v}_q, \vec{v}_d) = \frac{\vec{v}_q \cdot \vec{v}_d}{\|\vec{v}_q\|_2 \|\vec{v}_d\|_2} \quad (4)$$

3) Reciprocal Rank Fusion (RRF): Because BM25 yields unbounded positive scalars and Cosine Similarity yields bounded values $[-1, 1]$, standard score normalization is highly fragile. We bypass this scalar conflict via Reciprocal Rank Fusion (RRF) [13], synthesizing a unified ranking based purely on positional rank $r(d)$:

$$\text{RRF_Score}(d) = \sum_{r \in \{\text{Sparse}, \text{Dense}\}} \frac{1}{k_{rrf} + \text{rank}_r(d)} \quad (5)$$

where k_{rrf} is empirically constant at 60.

D. Generation Synthesis and Selective Abstention:

The current implementation enforces a strict structured output schema containing seven fields: `complaint_summary`, `predicted_issue`, `similar_cases`, `recommended_next_step`, `confidence`, `needs_human_review`, and `generation_mocked`.

The reported generation evaluation uses the repository’s deterministic mock provider rather than an external LLM. In mock mode, the complaint summary is constructed from the first portion of the source complaint, allowing the pipeline’s schema handling and control flow to be tested without an API dependency. Optional OpenAI and Anthropic provider implementations are present in the repository, but they were not used for the reported generation results.

For selective prediction, a fixed threshold $\tau = 0.60$ is applied directly to the classifier’s maximum raw softmax probability. The escalation parameter is defined as:

$$\text{needs_human_review} = \begin{cases} \text{True}, & \text{if } \max_c P(Y = c | X) < \tau, \\ \text{False}, & \text{if } \max_c P(Y = c | X) \geq \tau. \end{cases} \quad (6)$$

This mechanism is intended to route lower-confidence complaints to human review. The threshold was empirically set rather than learned from a calibration procedure, and the validation split is not used for threshold selection. Consequently, the reported method should be interpreted as confidence-based abstention rather than probability calibration. ECE measurement and validation-based threshold selection are left for future work.

Table 1: Comparative Accuracies of Classification Models

Classification Model	Global Accuracy	Macro-F1 Score	Weighted F1-Score
Multinomial Naive Bayes (TF-IDF)	80.85%	0.4153	0.7873
Logistic Regression (TF-IDF)	85.95%	0.6084	0.8468
LinearSVC (Platt Scaled, TF-IDF)	86.87%	0.6652	0.8608
Fine-Tuned DistilBERT (Proposed)	87.60%	0.6909	0.8709

Table 2: DistilBERT Performance by Financial Taxonomy Category

Category Name	Precision	Recall	F1-Score	Support
Credit Reporting	0.9213	0.9575	0.9390	4,938
Mortgage Servicing	0.9004	0.8490	0.8739	298
Bank Accounts / Services	0.7914	0.7851	0.7883	377
Debt Collection Interfaces	0.7830	0.6985	0.7384	806
Money Transfers	0.8208	0.7838	0.8018	222
Student Loans Systems	0.8073	0.7273	0.7652	121

IV. PERFORMANCE EVALUATION

The pipeline was empirically evaluated on the held-out 7,500-sample test set. Our assessment covers classification accuracy, retrieval precision, and the systemic safety of the abstention mechanisms.

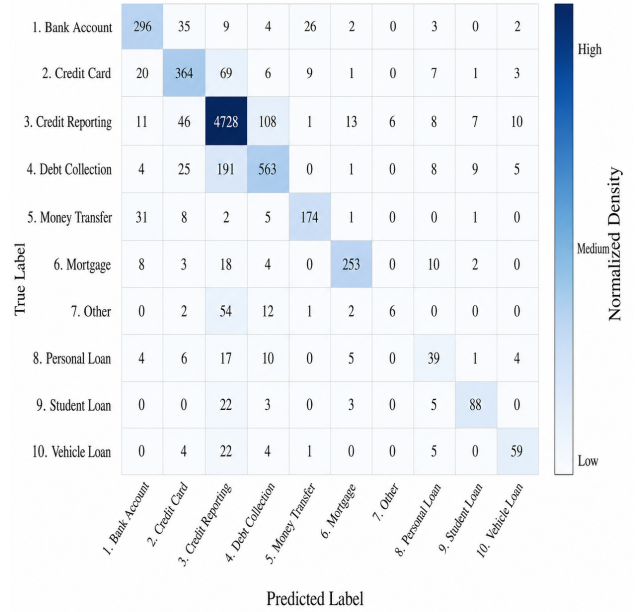
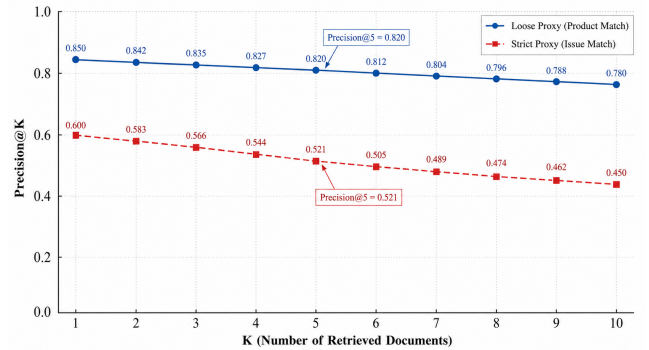
A. Classification Results: Table I shows that the contextual transformer performs better than the evaluated frequency-based baselines. While TF-IDF + Logistic Regression achieved a respectable 85.95% global accuracy, its Macro-F1 score (0.6084) revealed significant degradation on minority classes. The classical models leaned toward the majority class ("Credit Reporting"), contributing to high accuracy but weaker performance across several minority categories. DistilBERT effectively mitigated this variance, achieving 87.60% accuracy and raising the Macro-F1 to 0.6909.

Interpretation: DistilBERT improves Macro-F1 from 0.6084 for TF-IDF Logistic Regression to 0.6909, an increase of 0.0825 (8.25 percentage points). Because the test distribution remains highly imbalanced, Macro-F1 provides a more informative view of performance across classes than accuracy alone. The improvement is consistent with stronger minority-class performance, although the experiment does not isolate the causal contribution of contextual representation from other architectural differences.

Table II highlights the model’s precision and recall across highly skewed classes, demonstrating structural resilience even in complex, multi-faceted categories like Mortgage Servicing.

Interpretation: The Hybrid RRF retriever achieved a Precision@5 of 0.820 under the loose product-label match criterion and 0.521 under the stricter product-plus-issue criterion. These values are label-match proxy metrics rather than human relevance judgments. A retrieved complaint is counted as relevant when its `product_clean` label matches the query label, with the strict criterion additionally requiring the issue label to match.

The loose score therefore measures the retriever’s ability to return records from the same broad product category, rather

**Figure 2:** DistilBERT Classification Confusion Matrix Heatmap**Figure 3:** Retrieval Efficiency Scaling Graph (Precision@K).

than whether a human reviewer would consider the retrieved record a useful precedent. The lower strict score indicates that distinguishing closely related sub-issues remains more difficult. Human relevance annotation and qualitative usefulness evaluation are outside the scope of the current experiment.

For the reported evaluation configuration, retrieval queries are drawn from the held-out test split and the index is built from the training split. The current index-building interface defaults to the training split, although the code does not enforce train/test disjointness at runtime; reproduction should therefore verify that the retrieval index contains training records only.

C. Structured Generation and Abstention Control: The structured-generation path was tested in deterministic mock mode on a 30-sample batch. The mock provider does not invoke an external LLM; instead, it produces deterministic outputs to exercise the schema and pipeline control flow. The resulting outputs satisfied the required JSON schema for all 30 samples.

The selective-abstention rule was then applied to the classi-

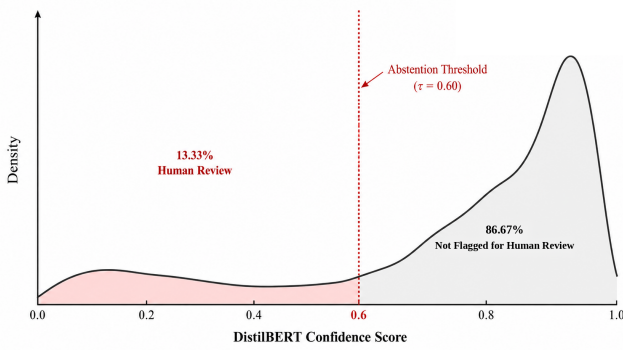


Figure 4: Raw Softmax Confidence Distribution and Fixed Abstention Threshold on the 30-Sample Mock Evaluation Batch.

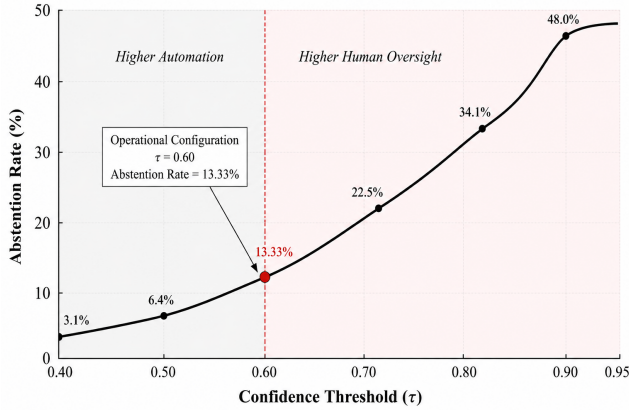


Figure 5: Threshold Sensitivity vs. Abstention Rate Curve.

fier’s raw softmax confidence. With $\tau = 0.60$, 4 of 30 samples were flagged for human review, corresponding to an observed abstention rate of 13.33%.

These results should be interpreted as structural and control-flow checks rather than evidence of generation quality, calibration, factuality, or RAG faithfulness. In particular, the mock summary is derived directly from the source complaint, so token-overlap grounding would be trivially high and is not treated as a meaningful generation-quality metric.

D. Limitations and Future Work: Several limitations bound the interpretation of the current results. First, retrieval relevance is evaluated using label-match proxies rather than human judgments. The loose Product Label Match criterion measures broad product-level agreement, while the stricter Product + Issue Match criterion provides a more demanding proxy for sub-issue retrieval. Neither establishes that a retrieved complaint would be useful to a human analyst.

Second, the generation evaluation is limited to a 30-sample deterministic mock run. The repository contains optional external LLM provider implementations, but no LLM generation quality, factuality, or end-to-end resolution-drafting evaluation is reported here. Any grounding metric computed in mock mode would also be dominated by the deterministic source-copy

behavior and is therefore excluded from interpretation.

Third, the confidence gate uses a fixed threshold on raw softmax probabilities. No temperature scaling, isotonic regression, ECE measurement, or validation-based threshold optimization is performed. The validation split is currently retained for future calibration and threshold-selection experiments.

Fourth, all reported experiments use a 50,000-record stratified sample. Although the repository supports chunked ingestion of the larger CFPB CSV, the reported model and retrieval metrics do not represent full-corpus experimentation.

Finally, the retrieval index builder defaults to the training split, while test complaints are used as evaluation queries. The current implementation does not enforce train/test index isolation at runtime, so future versions should add an explicit corpus/query disjointness check and guard against accidental self-retrieval.

Future work should therefore focus on calibrated selective prediction, human-annotated retrieval relevance, real LLM generation evaluation, and stronger end-to-end evaluation of resolution usefulness under human review.

V. CONCLUSION

This research presents a modular NLP pipeline for structured financial complaint analysis, combining redaction-aware preprocessing, transformer-based classification, hybrid sparse-dense retrieval, structured generation interfaces, and confidence-based human-review routing. Evaluation on a 50,000-record stratified sample shows that the fine-tuned DistilBERT classifier achieves 87.60% accuracy and a Macro-F1 of 0.6909, improving on the evaluated TF-IDF baselines while operating across a highly imbalanced class distribution.

The hybrid retrieval stage combines BM25 and dense MiniLM representations through Reciprocal Rank Fusion and achieves a Precision@5 of 0.820 under a product-label match proxy and 0.521 under a stricter product-plus-issue proxy. These values demonstrate useful product-level retrieval behavior but should not be interpreted as human-validated relevance.

The pipeline also exposes a fixed confidence threshold that routes lower-confidence predictions to human review. In a 30-sample deterministic mock-generation evaluation, 4 cases (13.33%) crossed the review threshold, and all outputs satisfied the required JSON schema. This experiment is a structural control check rather than evidence of LLM generation quality or calibrated probabilities.

Overall, the project demonstrates how classification, retrieval, and structured generation components can be separated into independently testable modules. The main remaining challenges are probability calibration, human relevance evaluation, real LLM generation assessment, stronger leakage guarantees, and validation of the complete workflow under realistic analyst feedback.

REFERENCES

- [1] Consumer Financial Protection Bureau. (2026). Consumer Complaint Database. [Online]. Available: <https://www.consumerfinance.gov/data-research/consumer-complaints/>

Table 3: Hybrid Retrieval Proxy Metrics (RRF, $k = 5$)

Proxy Match Type	Precision@5	Mean Reciprocal Rank (MRR)	NDCG@5
Product Label Match (Loose)	0.8200	0.8712	0.8839
Product + Issue Match (Strict)	0.5210	0.6264	0.6608

Table 4: Structured Generation and Selective Abstention Control Checks

Metric	Measured Value	Interpretation
Generation Mode	Deterministic mock	No external LLM used in the reported run
JSON Schema Validity	1.000 (100%)	Structural formatting check over 30 samples
Confidence Threshold	0.60	Fixed raw-softmax threshold for human-review routing
Abstention Rate	0.1333 (13.33%; 4/30)	Descriptive result from the 30-sample mock batch

- [2] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys (CSUR)*, vol. 55, no. 12, pp. 1-38, 2023.
- [3] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321-1330.
- [4] T. Joachims, "Text categorization with Support Vector Machines: Learning with many relevant features," in *European Conference on Machine Learning (ECML)*, 1998, pp. 137-142.
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [6] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171-4186.
- [7] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," arXiv preprint arXiv:1910.01108, 2019.
- [8] D. Araci, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," arXiv preprint arXiv:1908.10063, 2019.
- [9] P. Lewis, E. Perez, A. Piktus, E. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [10] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 3982-3992.
- [11] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535-547, 2019.
- [12] S. E. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333-389, 2009.
- [13] G. V. Cormack, C. L. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval*, 2009, pp. 758-759.
- [14] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proceedings of the 22nd international conference on machine learning (ICML)*, 2005, pp. 625-632.
- [15] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [16] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Advances in large margin classifiers*, vol. 10, no. 3, pp. 61-74, 1999.
- [17] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. draft, 2023.

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